Shubham Gogri

Machine Learning Engineer | Production ML & MLOps | MSc AI (Distinction)

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Profile

Machine Learning Engineer building and operating production ML systems end-to-end — from research and feature engineering through deployment, drift monitoring, and data-driven retraining. Currently at Cloudbeds, owning demand-forecasting and price-optimisation models that turn multi-source consumer and market data into commercial recommendations across 20,000+ properties.

Technical Skills

ML & Data: PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy, PySpark, SciPy, SHAP, Pydantic, Pandera, pytest, Snowflake, PostgreSQL, Redshift, Streamlit, Plotly Dash

Cloud & MLOps: AWS (ECS, Lambda, S3, SageMaker, Batch, ECR, EKS/Kubernetes, Feature Store), Apache Airflow, Docker, Terraform (IaC), Git, GitHub Actions, MLflow, DataDog, CI/CD, Kafka

Languages & Tools: Python, Java, SQL, FastAPI, Flask, Tableau, Power BI, Weights & Biases

Experience

Machine Learning Engineer — Cloudbeds London, UK | Jul 2024 – Present

- Authored ML systems-design proposals presented directly to the Engineering VP, defining the architecture for onboarding thousands of hotels onto the predictive pricing platform as a commercial scaling strategy.
- Owned an end-to-end drift-detection system for deployed time-series forecasting models — scoped, built, and operated it — monitoring distribution shift and performance degradation beyond standard metrics to safeguard production reliability and trigger retraining.
- Built deterministic, tree-based time-series demand forecasting models powering a dynamic pricing engine, with **SHAP** attributions for per-property explainability and revenue-manager trust.
- Engineered forecasting features from multiple distinct sources — booking trends, events, competitor and hotel rates, weather forecasts, and air-traffic disruption data — building the multi-source signal layer underpinning demand prediction.
- Led migration of the Kafka-streaming feature pipeline from Pandas to PySpark, **cutting computation time 40%** while processing **50M+ daily booking events**, with built-in distribution-shift monitoring.
- Determined optimal retraining cadence via non-parametric hypothesis testing (Wilcoxon, Kruskal–Wallis) and OLS with interaction terms, modelling error as a function of model age — **establishing that monthly retraining with fixed hyperparameters outperforms weekly**, reducing prediction volatility.
- Built FastAPI model-serving microservices on Kubernetes with Terraform IaC and Airflow-orchestrated batch inference, developed in a modular pattern with QA gates and unit, integration, and regression tests — delivered in Scrum sprints with product managers.
- Built an experimental validation framework that surfaced edge-case failures in **30% of properties**, enabling a segmented commercial rollout across **20,000+ hotels** and de-risking a costly full deployment.
- Drove early-stage research with clients' revenue managers to refine prediction accuracy and optimisation strategy, defining automated smoke tests to validate recommendations before release.
- Introduced a multi-modal clustering solution feeding LLM-derived embeddings (OpenAI, Llama, JinaAI) into variational autoencoders to project demand curves for cold-start hotels with limited history.

Technologies: Python, PyTorch, XGBoost, scikit-learn, PySpark, Kafka, FastAPI, Airflow, Kubernetes, Docker, Terraform, MLflow, DataDog, SHAP, Snowflake, pytest

Data Scientist — Qvantia Remote, UK | 1-month contract, MONTH YYYY

- Built a supervised anomaly-detection pipeline (Random Forest, XGBoost) on multivariate sensor streams, identifying production faults across manufacturing systems.
- Applied PCA and LDA to high-dimensional sensor inputs to address multicollinearity, improving model robustness and interpretability.

Education

University of Surrey — MSc Artificial Intelligence, Distinction Feb 2024

Projects

Deep Learning for Image Correction in Sparse-View CT PyTorch/TensorFlow, Computer Vision, Medical Imaging

- Reconstructed sparse-view CT scans with a Pix2Pix GAN using custom perceptual and Wasserstein losses, achieving **PSNR 76** and **SSIM 1.0** to reduce radiation-dose requirements.
- Designed containerised training pipelines with TensorBoard monitoring, deployed on a university HPC cluster via Docker for scalable, reproducible experimentation.

Emotion Classification NLP, Deep Learning, MLOps

- Classified 27 emotion labels in Reddit comments (GoEmotions) by training DistilBERT, Bi-LSTM, and CNN-LSTM models, tracking experiments and hyperparameter trade-offs via Weights & Biases.
- Deployed the fine-grained classifier as a Gradio web app hosted on Hugging Face Spaces.

Publications & Recognition

- *GAN-based reconstruction for sparse-view CT.* Oral presentation, International Conference on Machine Learning in Medical Imaging (ICMLMI), 2023.
- **Top 3 — Best Dissertation Project**, Electronic Engineering Industrial Advisory Board MSc Project Prize, University of Surrey (department-wide, industry-panel judged).
- **2nd prize, Generative AI Hackathon**; *Excellence in AI Collaboration* — Surrey SetSquared entrepreneurship programmes (IKEEP, ITeK).